Stronger spin-one gap bounds from the same finite certificates

Computational research note

8 October 2026

The exact Julia scalar calculations and the temperature-lifting argument below, combined with the analytic transfer and bootstrap statements in *The periodic spin-one Haldane gap*, give

$$\boxed{\gamma_L > 0.072901771 \quad \text{for every even } L \geq 36.}$$

Here $H_L = \sum_{j=1}^{L} \mathbf{S}_j \cdot \mathbf{S}_{j+1}$, $\mathbf{S}^2 = 2I$, and the bond coupling is one. The ground state is unique in this range. A simpler temperature-lifting choice gives the readable statement $\gamma_L > 7/100 = 0.07$ in the same range. The calculations reuse the existing trial vector, thermal contractions, shift, and polynomial filters. The new bound combines the stronger thermal data with the length-120 trial and raises inverse temperature before reducing the information to two purities.

Calculation	Gap lower bound	Even lengths covered
Published source 1	$\log(20)/784 \simeq 0.003821087$	$L \geq 2304$
Published source 2	$(4/105)\log(80/79) \simeq 0.000479192$	$L \geq 60$
Improved source 1	0.016978269	$L \geq 36$
Improved source 2	0.011977901	$L \geq 60$
Temperature lifting, simple	$0.070450002 > 7/100$	$L \geq 36$
Temperature lifting, optimized	0.072901771	$L \geq 36$

Source 1 uses the eighteen thermal values at lengths 4–12 and the length-72 trial certificate. Source 2 uses only nine thermal values at lengths 6, 8, and 10 and the length-120 trial certificate. Thus source 2 remains useful as the less expensive input set, even though the improved source-1 conclusion is stronger. The temperature-lifting calculation uses the eighteen source-1 thermal values and the length-120 trial; the original length-72 seed covers the early lengths. All these finite inputs were already reproduced. No global optimum is claimed.

1 Comparison with the numerical bulk gap

For the same convention $J = 1$, the large-system numerical estimates are

$$\Delta_{S=1} \simeq 0.41048(6), \qquad \Delta_{S=2} \simeq 0.08917(4).$$

These are the estimates in Table I of S. Todo and K. Kato, *Cluster Algorithms for General-S Quantum Spin Systems*, Phys. Rev. Lett. **87**, 047203 (2001). That table reports spatial correlation lengths 6.0153(3) and 49.49(1), respectively. The quoted numerical uncertainties are not rigorous spectral enclosures. The spin-one DMRG estimate 0.41050(2) of White and Huse, Phys. Rev. B **48**, 3844 (1993), is consistent with this scale.

The temperature-lifting lower bound is approximately 17.76% of the numerical bulk gap, compared with 4.14% for our earlier 0.016978269 bound and 0.93% for the published log(20)/784 bound. It improves our earlier bound by about 4.29 and the published bound by about 19.1, while remaining about 5.63 times below the numerical value. These statements compare a uniform bound on finite periodic chains with the numerical bulk benchmark; we have not independently reproduced the quoted thermodynamic-limit gap.

The selected quantity $A_j = (1-p_j)^2(1-q_j)$ below is a dimensionless *purity lower bound*, not a gap estimate. Along the successful bootstrap, where both length and inverse temperature grow as 2^j, the controlled defects tend to zero and $A_j \to 1$. The gap information is in the rate, through $\log[A_j/(1-A_j)]/b_j$, rather than the limiting value 1 itself. This is not a statement that purity tends to one at fixed temperature as length grows.

2 Input scope and provenance

The immutable upstream source is the OpenAI math repository, commit `adc7f1241b42e322a6451854ab7e4b4c146` directory `The-periodic-spin-one-Haldane-gap-September-24-2026` under `preprints`. The relevant arguments are in `build/initialization.tex` and `build/bootstrap.tex`. The previously completed Julia reproduction checks all finite thermal and trial integers against the upstream evidence. The new scalar calculation checks that reproduction's implementation and component hashes before using the input data. It does not describe reusing those outputs as fresh thermal contraction.

Let $a = 700741/500000$ and

$$Z_n(b) = \operatorname{Tr} e^{-b(H_n+naI)}, \qquad S(n,b) = \frac{Z_{2n}(b)}{Z_n(b)^2}, \qquad T(n,b) = \frac{Z_n(2b)}{Z_n(b)^2}.$$

The paper's transfer theorem supplies the real spatial eigenvalue lists and their sector multiplicities. Its length-72 and length-120 trial inequalities give $Z_{72}(b) > 1$ and $Z_{120}(b) > 1$ for every $b > 0$.

3 A compact improved source-1 initialization

Retain the original filters and their original error allowances:

$$\begin{aligned} F_J(x) &= x^4(7 + 17x - 280x^2 - 185x^3 + 1000x^4)^2, \\ F_N(x) &= x^4(12 - 394x^2 + 1000x^4)^2, \\ B_J &= 318604.54848175, \qquad B_N = 48169.880699. \end{aligned}$$

Every terminating decimal in a defining formula here denotes an exact rational. Choose

$$U = 1.001351, \quad V = 0.871256, \quad \ell = 0.999997, \qquad m_J = 1.0657205, \quad m_N = 0.2783155.$$

The verifier checks $F_J(\pm U) > B_J$ and $F_N(\pm V) > B_N$. It expands each $F(\pm(v+y))$ and checks every nonconstant coefficient is nonnegative. This verifies both entire exterior rays, so the paper's polynomial-filter lemma gives $|\lambda| < U$ in J and $|\lambda| < V$ in N.

Write

$$\mathcal{C}(m,h,r) = \lfloor m/h \rfloor h^r + (m - \lfloor m/h \rfloor h)^r, \qquad B(k) = 3\mathcal{C}(m_N, V^{12}, k/12).$$

The same capped-mass argument as in the paper now verifies

$$\mathcal{C}(m_J, \ell^{12}, 6) + B(72) < 1, \qquad \ell^{12} < m_J < 2\ell^{12}.$$

Since $Z_{72}(49/4) > 1$, one spatial eigenvalue in J has modulus greater than ℓ. Removing it leaves twelfth-moment mass at most $m_J - \ell^{12}$. Hence, for $k = 36, 72$,

$$R_A(k) = B(k) + (m_J - \ell^{12})^{k/12}$$

bounds the residual kth moment. At $(n, b) = (36, 49/4)$ we therefore have

$$1 - S(36, b) \le p_0 = 1 - (1 + R_A(36)/\ell^{36})^{-2},$$
$$1 - T(72, b) \le q_0 = 1 - (U^{72} + R_A(72))^{-2}.$$

Their numerical values are about 0.0449638076 and 0.176903757. These initial bounds need not satisfy the paper's common-defect bootstrap hypotheses. The coupled update permits unequal defects.

4 Cover the early lengths before applying the uniform theorem

The following finite-prefix version is a direct consequence of the paper's coupled update, purity interpolation, and uniform bootstrap proposition.

Proposition 1. *Suppose p_j, q_j are valid upper bounds on the spatial and temporal defects at $(n_j, b_j) = (2^j n_0, 2^j b_0)$ for $0 \le j \le K$. Put $A_j = (1 - p_j)^2(1 - q_j)$ and suppose $A_j > 1/2$ for $j < K$. Suppose $u = \max(p_K, q_K)$ and C satisfy*

$$0 < u < 1/6, \quad G(u) := \tfrac{1}{2}\big((1-u)^{-1} + (1-u)^{-2} + (1-u)^{-3}\big)^2 \le C, \quad Cu < 1, \quad C(1-3u) > 3.$$

If g is strictly smaller than every member of

$$\left\{ \frac{\log(A_j/(1-A_j))}{b_j} : 0 \le j < K \right\} \cup \left\{ \frac{\log(1/(Cu))}{b_K} \right\},$$

then the ground state is unique and $\gamma_L > g$ for every even $L \ge n_0$.

Proof. For $n_j \le L \le 2n_j$, the interpolation inequality gives $T(L, b_j) \ge A_j^{L/(2n_j)} \ge A_j > 1/2$. The largest physical Gibbs weight w_0 is at least this purity, so it is unique. For a weight w_1 at the first excited energy,

$$e^{-b_j \gamma_L} = w_1/w_0 \le (1 - w_0)/w_0 \le (1 - A_j)/A_j.$$

This proves the required estimate on each finite interval. The original uniform bootstrap proposition covers every even $L \ge n_K$. The intervals $[n_j, 2n_j]$ for $j < K$ and the last half-line cover all even lengths at least n_0. □

Use the original update $f(u) = u^2/[2(1-u)^2]$, retaining the two defects:

$$p_{j+1} = \big\lceil f(1 - (1 - p_j)^2(1 - q_j)) \big\rceil_{100},$$
$$q_{j+1} = \big\lceil f(1 - (1 - q_j)^2(1 - p_{j+1})) \big\rceil_{100}, \qquad \lceil x \rceil_{100} = 10^{-100} \lceil 10^{100} x \rceil.$$

For the compact constants, take $K = 9$ and $C = 5$. The verifier checks all uniform-theorem hypotheses and all intervening length intervals. The following entries are explanatory approximations; full rational bounds are in the certificate.

j	p_j	q_j	$\log(A_j/(1-A_j))/b_j$
0	0.0449638076	0.176903757	0.0900056767
1	0.0551175965	0.158002808	0.0452204114
2	0.0545331044	0.120970423	0.0265228864
3	0.0371649174	0.0592125617	0.0195936555
4	0.0107429704	0.0100975890	0.0175199506
5	0.0005203714	0.0002212682	0.0170259785
6	$7.97715\,10^{-7}$	$9.83379\,10^{-8}$	0.0169496858
7	$1.43443\,10^{-12}$	$1.93410\,10^{-14}$	0.0169454007
8	$4.17084\,10^{-24}$	$7.48147\,10^{-28}$	0.0169453721

At $j = 9$, $n_9 = 18432$, $b_9 = 6272$, and $u \simeq 3.47981322\,10^{-47}$. The uniform tail bound exceeds $0.016799278 > 1/60$. All earlier interval bounds exceed $1/60$ as well. This proves the compact claim.

5 Tighter constants and the less expensive branch

Exact bisection on a grid of spacing 10^{-12} gives valid source-1 choices

$$U = 1.001350644012, \quad V = 0.871255704653, \quad \ell = 0.999997937629.$$

Eleven coupled updates, upward rounding at spacing 10^{-512}, and an outward-rounded $C \geq G(u)$ then establish the earlier improved bound 0.016978269. The very fine rounding avoids a floor in the rapidly squaring defects; it concerns only scalar integers, not tensor dimensions.

For source 2 the script preserves the filters $x^6(-2+10x^2)^2$ and $x^6(-25+100x^2)^2$, the thermal error 10^{-5}, the moment degrees, and the temperature multiplier $5/2$. It sharpens the two cutoffs to $U = 1.002408155316$, $V = 0.886411420539$, uses the sector-center-plus-error moment bounds directly, and obtains rational upper bounds

$$D = 1.445289213010, \quad E = 1.198064585836, \quad R_q = 0.064761703869.$$

As in the original proof, $D^2 > m^5$, $E^2 > p^5$, and $R_q^2 > (q-1)^5$. Set $\tau = (D+11E)/12$ and $\tau_0 = (D+3E)/4$. The original four-vertex residual bound $R(t)$ is repeatedly used to upgrade the dominant weight whenever $s^4 + R(t)^2 < 1$. The final lower bound is $t = 0.998895181835$; the resulting initial defects are approximately 0.1209336963 and 0.1179460383. Applying the same finite-prefix proposition gives $\gamma_L > 0.011977901$ for every even $L \geq 60$.

For comparison, merely retaining the original exact seeds and improving the bootstrap gives 0.004710839 at $L \geq 36$ from source 1 and 0.006924382 at $L \geq 60$ from source 2. Thus both tighter initialization and better use of the coupled recurrence contribute to the improvement.

6 Temperature lifting before the purity bootstrap

The following recombination of the same finite inputs gives a larger gain. It generalizes the paper's source-2 temperature argument to the stronger source-1 moments, retaining the threefold sector multiplicity at every required moment order.

Use the tighter rational U, V, m_J, m_N above and put $b_0 = 49/4$. For every even $k \geq 12$, capped masses give

$$\begin{aligned} \mathcal{J}(k) &= U^k + (m_J - U^{12})^{k/12}, \\ \mathcal{N}(k) &= V^k + (m_N - V^{12})^{k/12}, \\ Z_k(b_0) \leq M_k &:= \mathcal{J}(k) + 3\mathcal{N}(k), \qquad Z_k(b_0, P) \leq \mathcal{J}(k). \end{aligned}$$

Indeed $U^{12} < m_J < 2U^{12}$ and $V^{12} < m_N < 2V^{12}$. The last inequality uses $Z_k(b_0, P) = J_k - N_k$ and $N_k \geq 0$.

Choose $r \geq 1$ and an even $12 \leq k \leq 60$. Because each twisted partition function is itself the Gibbs trace of a self-adjoint Hamiltonian, raising its positive Boltzmann factors to the power r gives

$$Z_k(rb_0) \leq D := M_k^r, \qquad Z_k(rb_0, P) \leq E := \mathcal{J}(k)^r.$$

Outward rational upper bounds may replace D, E. In the new spatial operator at $b = rb_0$, set $w_i = |\lambda_i(b)|^k$ and

$$h = 60/k \geq 1, \qquad \tau = (D + 3E)/4.$$

Then $\sum_i w_i \leq D$, the J-sector mass is at most τ, and the existing length-120 trial gives

$$\sum_i w_i^{2h} = Z_{120}(b) > 1.$$

Choose a rational $t > 0$ with $Dt^{2h-1} < 1$ and $3t > D$. Some weight $w_0 > t$ exists, since otherwise the last sum is at most $Dt^{2h-1} < 1$. This weight belongs to J: every N weight occurs three times, which would contradict the total mass bound. After removing w_0, let x be the remaining J mass. Then $0 \leq x \leq \tau - t$, while the full remaining N mass is at most $D - t - x$. For every $s \geq 1$, the residual sth moment is at most

$$R_s(t) = \max\left\{3\left(\frac{D-t}{3}\right)^s, (\tau - t)^s + 3\left(\frac{D-\tau}{3}\right)^s\right\}.$$

To prove this, bound each sector list by the sth power of its total mass. The three identical N lists contribute at most $3[(D - t - x)/3]^s$. The resulting convex function of x attains its maximum at one of the two endpoints. All quantities are nonnegative because $t \leq \tau \leq D$.

We may repeatedly replace t by a larger rational t' whenever

$$(t')^{2h} + R_{2h}(t) < 1.$$

If $w_0 \leq t'$, this inequality would contradict $Z_{120}(b) > 1$. Using R_{2h} directly preserves more multiplicity information than squaring an upper bound on R_h. Consequently a spatial seed at length 60 is

$$S(60, b) \geq \left(1 + \frac{R_h(t)}{t^h}\right)^{-2}.$$

The temporal seed uses the same trial in the physical spectrum. Write $Q = M_{120} > 1$. At b_0, the ground Boltzmann factor $z_0 > 1$ and the total mass is at most Q. Hence

$$\sum_{i>0} \frac{z_i}{z_0} \leq Q - 1, \qquad \sum_{i>0} \left(\frac{z_i}{z_0}\right)^r \leq (Q-1)^r.$$

This yields the second seed without a new contraction:

$$T(120, b) \geq \left(1 + (Q-1)^r\right)^{-2}.$$

The spatial weights w_i and physical Boltzmann factors z_i play different roles; the proof uses the original transfer theorem to connect their partition functions.

k	r	$1 - S(60, rb_0)$ upper	$1 - T(120, rb_0)$ upper	Gap bound
30	2	0.0307608679	0.0590740709	0.070450002
28	17/8	0.0305302052	0.0479556776	0.072901771

The displayed defect bounds are rounded upward for readability; the calculation uses full rational enclosures. Ten coupled updates use an upward rounding grid 10^{-2048}, keeping its floor below the relevant defects, followed by the uniform tail theorem. This covers even $L \geq 60$. The earlier source-1 seed gives an interval bound greater than 0.09 on every even $36 \leq L \leq 72$. Combining these overlapping ranges establishes both displayed bounds for every even $L \geq 36$. The simpler row just doubles inverse temperature and uses $h = 2$; its residual estimates therefore involve only second and fourth powers. The numerical discovery script scans even k from 12 to 60 and r from 1 to 3 in increments of 1/40. Its output is not a certificate; the two selected parameter choices are replayed through the rational verifier.

7 Reproduction and the arithmetic trust boundary

Run from the repository root:

```
julia --startup-file=no --project=models/spin1 \
  models/spin1/experiments/optimize_bounds.jl
julia --startup-file=no --project=models/spin1 \
  models/spin1/experiments/test_bounds.jl
julia --startup-file=no --project=models/spin1 \
  models/spin1/experiments/temperature_lift.jl
julia --startup-file=no --project=models/spin1 \
  models/spin1/experiments/test_temperature_lift.jl
```

The output `models/spin1/experiments/bounds.json` contains the complete initialization parameters, every rational defect pair, interval coverage, the tail-theorem parameters, logarithm enclosures, and source hashes. All accepting comparisons use integers or rationals. Displayed floating-point numbers are diagnostic only. The new `temperature-lift.json` records the lifted moment bounds, dominant-weight refinements, both bootstrap chains, and the early-length bridge, together with script and input hashes. Fractional powers are enclosed using integer roots: for $x \geq 0$ and $r = a/b \geq 0$, the verifier checks rational endpoints l, u through $l^b \leq x^a \leq u^b$. No floating-point comparison accepts a certificate.

To enclose a logarithm, range-reduce $x = 2^k y$ with $1 \leq y < 2$, round y outward to a rational grid, and use $z = (y-1)/(y+1)$ in

$$\log y = 2 \sum_{j=0}^{M-1} \frac{z^{2j+1}}{2j+1} + E_M, \qquad 0 \leq E_M \leq \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

Here $M = 40$ and $0 \le z \le 1/3$; $\log 2$ is treated by the same formula. The final rational claim is strictly below all computed lower endpoints.

The tests independently rearrange the coupled updates, verify continuous coverage of lengths, check the seed inequalities and reject unsafe cutoffs, rebuild all 27 input thermal enclosures, and check both variational signs. They also use high-precision floating-point logarithms as an independent numerical control, not as certificate authority. This is exact finite arithmetic combined with the paper's analytic arguments; it is not a Lean formalization of the transfer theorem or of the infinite induction. The physical trial still has bond dimension 26.

8 Spin-two variational certificates and the remaining thermal input

The spin-two extension uses $\mathbf{S}^2 = 6I$ and the same coupling $J = 1$. The scalar coupled recurrence and the finite-prefix argument above apply once its model-specific thermal and variational inputs have been supplied. The smaller numerical gap and longer correlation length make those inputs more demanding; the spin-one gap constant cannot simply be rescaled.

The first Julia search used virtual space

$$V_0 \oplus 2V_1 \oplus 2V_2 \oplus V_3, \qquad D = 24,$$

with 29 real reduced coefficients. Exact highest-weight equations construct integer polynomial tensors. After rounding the numerical coefficients to integers, the physical matrices have the form

$$A_m = \sqrt{\binom{4}{2+m}}\, P_m, \qquad P_m \in \mathbb{Z}^{24\times 24}, \quad -2 \le m \le 2.$$

The square roots cancel in the norm and bond-energy contractions, leaving integer transfer blocks. Their largest dimension is 106. The finite certificate verifies

$$\frac{\langle \psi_N, H_N \psi_N \rangle}{N \langle \psi_N, \psi_N \rangle} < -4.7595 \qquad (N = 128, 256, 512).$$

The corresponding energy densities, shown only as approximations, are -4.7595998798, -4.7595993033, and -4.7595993033. Thus $E_0(N) < -4.7595N$ and the shifted partition function exceeds one at each of these lengths for every positive inverse temperature. Direct four-site integer amplitudes agree with the transfer contraction; an independent ITensors MPO/MPS calculation agrees in total energy within 7.2×10^{-15}.

The spin-two numerical bulk energy density in Todo–Kato is $-4.761249(6)$, so this first trial is still roughly 0.00165 per bond above that reference. Its energy certificate does not certify a gap. An improved search uses analytic derivatives of both transfer fixed points and factors the energy channel into small multiplicity matrices. At virtual dimension 137, quantization to integer coefficients followed by exact representation checks and outward-rounded Arb contractions gives the strict finite trial-energy intervals

N	Lower endpoint	Upper endpoint
128	-4.761268026389	-4.761268026385
256	-4.761228827807	-4.761228827803
512	-4.761228489623	-4.761228489619

These enclose the normalized *trial* energy, not the exact ground energy. In particular, $E_0(N) < -4.76122N$ at all three lengths. The largest reduced matrix is 529×529, in place of the full 18769×18769 transfer matrix. The verifier checks exact covariance, complete irreducible bases, and every highest-weight image before evaluating the long-chain traces. The full run took about 489 seconds. The integer witness and strict rational intervals are saved in `spin2/results/trial-137-certificate.json`. A further numerical search at dimension 197 gives energy per bond -4.7612412068. Enlarging the multiplicity spaces to dimension 295 and using a regularized transfer-metric preconditioner gives -4.7612463445. These two values are numerical search results. The $D = 295$ coefficients are saved in `spin2/results/trial-295-refined.json`.

The larger verifier uses exact tensor covariance and complete irreducible bases to determine the reduced channels from highest-weight coefficients. Schur's lemma determines the other weights. A further exact involution, $X \mapsto UX^{\mathsf{T}}U^{\mathsf{T}}$ with U a spin rotation by π, splits these matrices into two smaller blocks. The implementation checks commutation of both reduced channels with this involution over the rationals. Independent full-image checks and the split and unsplit contractions agree in all 92 certificate tests. Sparse ladder arithmetic reduced the measured $D = 295$ matrix-construction time from about 390 to 62 seconds; this comparison does not include the subsequent interval matrix powers.

The larger finite-energy certificate has now completed. Its strict trial-energy intervals are

N	Lower endpoint	Upper endpoint
128	-4.761288829076	-4.761288829072
256	-4.761247250800	-4.761247250796
512	-4.761246333989	-4.761246333985
1024	-4.761246333503	-4.761246333499

Thus $E_0(N) < -4.7612463N$ at all four lengths. The largest arithmetic block has dimension 978, reduced from 1921 by the verified involution. The complete run took about 549 seconds with four FLINT threads. The integer witness, source hashes, and interval results are in `spin2/results/trial-295-symmetr`

No spin-two finite-temperature purity seed has yet been certified. The existing periodic DMRG pilot gives gaps near 0.54557 at $L = 8$ and 0.3109 at $L = 16$; the latter has appreciable residuals. These short-chain numbers should not be used as estimates of the 0.08917 bulk gap.

The scalar screening has also been relaxed from requiring both purities to exceed 0.99. Exact recurrence checks admit, for example, target purities $(S, T) \geq (0.95, 0.83)$. If such targets were realized, the necessary energy screen would be

$$\frac{\log[T/(1-T)]}{\Delta_{2n}} \leq \beta \leq \frac{-\log(ST)}{E_0(2n) - 2E_0(n)},$$

for positive denominator. This permits the energy-mismatch-to-gap ratio to be as large as 0.14986, versus 0.00437435 for the old 0.99 targets. It is only a necessary screen; neither target pair has been established for spin two, and the current small-chain pilot fails even the relaxed screen.

The scripts `spin2/scripts/certify_trial.jl` and `spin2/scripts/bootstrap_screen.jl` reproduce these two steps. The full integer trial parameters, norms, energy numerators, and block contributions are retained in `spin2/results/trial-certificate.json`.

8.1 A conditional criterion using one thermal upper bound

The improved trial permits a useful simplification of the thermal task. Write $\widehat{Z}_L(b) = \operatorname{Tr} e^{-b(H_L + aL)}$; scalar shifts preserve the purities and rescale the self-adjoint spatial transfer operator.

Proposition 2 (One-trace temperature lifting)**.** *Let $n \geq 4$ be even, let $h, r \geq 1$ be integers, and suppose*

$$\widehat{Z}_n(\beta) \leq U, \qquad E_0(2hn) < -2hna, \qquad U > 1.$$

Set $D = U^r$ and $Q = U^{2h}$. Choose $t_0 > 0$ satisfying $Dt_0^{2h-1} < 1$ and $2t_0 > D$. One may take $t = t_0$, or any $t \in [t_0, 1]$ satisfying $t^{2h} + (D-t)^{2h} < 1$. Then

$$S(hn, r\beta) \geq \left[1 + \left(\frac{D-t}{t}\right)^h\right]^{-2}, \qquad T(2hn, r\beta) \geq [1 + (Q-1)^r]^{-2}.$$

Proof. The physical power inequality gives $\widehat{Z}_n(r\beta) \leq U^r = D$. For the shifted spatial transfer eigenvalues at $r\beta$, put $w_i = |\lambda_i|^n$. Thus $\sum_i w_i \leq D$ and $\sum_i w_i^{2h} = \widehat{Z}_{2hn}(r\beta) > 1$ by the variational hypothesis. If $\max_i w_i \leq t_0$, then the latter sum is at most $Dt_0^{2h-1} < 1$, a contradiction. The largest weight therefore exceeds $t_0 > D/2$. On this interval, $x^{2h} + (D-x)^{2h}$ is increasing. Bounding the remaining $2h$-moment by $(D-x)^{2h}$ proves that the largest weight also exceeds any admissible t. The remaining h-moment is at most $(D-t)^h$, proving the spatial estimate.

At inverse temperature β, the spatial moment inequality gives $\widehat{Z}_{2hn}(\beta) \leq U^{2h} = Q$. Its ground-state Boltzmann weight exceeds one, so the sum of all excited weights relative to the ground weight is less than $Q - 1$. Raising inverse temperature by the integer r bounds this sum by $(Q-1)^r$. The squared ground-state probability alone gives the stated thermal-purity bound. □

For example, with $n = 128$, $\beta = 32$, $h = 2$, $r = 3$, and $U = 1.08$, exact rational scalar checks give the *conditional* conclusion

$$\Delta_L > 0.006039844 \qquad \text{for every even } L \geq 256.$$

The length-512 variational hypothesis is supplied by the certificate above with $a = 4.76122$. However, at this shift the length-128 trial already gives

$$\widehat{Z}_{128}(32) > e^{32 \cdot 128(4.761268026385 - 4.76122)} > 1.1967 > 1.08.$$

Thus this illustrative thermal target is **ruled out**, rather than merely unverified. Larger thermal lengths and more accurate trial shifts must be explored. The proposition supplies a family of conditional tests, not a spin-two gap proof. The script `spin2/scripts/one_trace_target.jl` and its 63 tests check the conditional scalar argument.

The new length-1024 certificate supplies the trial hypothesis for

$$n = 256, \qquad \beta = 64, \qquad h = 2, \qquad r = 3, \qquad a = 4.7612463.$$

At these parameters, the still-unproved thermal inequality $\widehat{Z}_{256}(64) \leq 1.08$ would imply

$$\Delta_L > 0.003019921 \qquad \text{for every even } L \geq 512.$$

This target passes the preceding variational rejection screen; that necessary check does not establish the thermal inequality. The exact conditional calculation is saved in `spin2/results/one-trace-current-targets`

8.2 Thermal upper bounds and compression

Split the even ring into disjoint odd and even bond layers, $H = A + B$. The Araki–Lieb–Thirring inequality and the product-formula limit imply

$$Z_N(\beta) \leq \operatorname{Tr}\left(e^{-\beta A/m} e^{-\beta B/m}\right)^m, \qquad m \in \mathbb{N}.$$

Thus a certified contraction of the right side needs no separate estimate of the Trotter error. A Julia reference implementation uses exact rational two-site spectral projectors and outward-rounded Arb products in every magnetization sector. Its 42 checks include the independently known four-site spectrum. It currently treats only reference sizes.

After the Marshall transformation the gates are nonnegative entrywise. A candidate compression can therefore be replaced by rational nonnegative factors satisfying $M \le UV$ entrywise, checked exactly; subsequent positive contractions preserve this inequality. A four-site prototype bounds the thermal amplitude in this way. At $\beta = 4$, its upper bound exceeds the uncompressed Trotter trace by a factor about 1.00844 at bond dimension 569 (the full cut has dimension 625). This establishes a small-system compression certificate, but not yet an efficient scheme for the large thermal traces needed above.

Repeated compression is more demanding than this single-compression test. Starting at $\beta = 1/8$ and making five squarings, with compression at the initial stage and after each squaring, gives the following four-site results at $\beta = 4$, with 64 Trotter steps. Each entry is the certified upper trace divided by the uncompressed Trotter trace; displayed ratios are approximate.

Compression certificate	Represented rank bound	Upper/reference
Entrywise positive majorant	569	3.25495415
Hilbert–Schmidt error	564	1.00841069
Spectral majorant	565	1.00388214

The full four-site cut has dimension 625. These results do not establish the cost or accuracy of a long-chain contraction.

For the second row, let A be the exact Hermitian thermal amplitude and B its Hermitian approximation. If $\|A-B\|_{\mathrm{F}} \le \varepsilon$ and $\|B\| \le b$, and a new approximation C satisfies $\|B^2-C\|_{\mathrm{F}} \le \delta$, then

$$\|A^2 - C\|_{\mathrm{F}} \le \delta + \varepsilon(2b + \varepsilon).$$

Indeed, writing $E = A - B$ gives $A^2 - B^2 = BE + EB + E^2$ and applying the operator/Hilbert–Schmidt norm inequalities proves the claim. Consequently

$$\operatorname{Tr} A^2 \le (\|B\|_{\mathrm{F}} + \varepsilon)^2.$$

The verifier includes outward-rounded matrix-product errors in δ. SVD truncation respects charge and transpose parity; rational rounding preserves Hermiticity without doubling the represented rank.

The third row uses a different induction. Suppose $M \ge 0$ and its ordered eigenvalues dominate those of $A \ge 0$. Given Hermitian C and $\|M^2 - C\| \le \delta$, put $M' = C + \delta I$. Then $M' \ge M^2 \ge 0$. The min–max principle gives

$$\lambda_i(M') \ge \lambda_i(M^2) = \lambda_i(M)^2 \ge \lambda_i(A)^2 = \lambda_i(A^2).$$

This preserves spectral domination at every squaring. The final $\operatorname{Tr}[(M')^2]$ therefore bounds the desired trace. Adding the identity costs at most one additional Schmidt term.

Both norm certificates currently obtain an operator-norm bound by enumerating the four-site magnetization sectors. In the rational polynomial gauge, for a matrix R similar to a Hermitian matrix and any strictly positive rational vector v,

$$\|R\|_{\mathrm{physical}} \le \max_i \frac{(|R|v)_i}{v_i},$$

where $|R|$ means entrywise absolute value. The candidate v is numerical, but the displayed upper bound is checked rationally. Replacing this dense step by a sufficiently sharp long-chain norm witness remains an open computational task. The results are saved in `repeated-positive-compression.json`, `repeated-l2-symmetric.json`, and `loewner-compression.json` under `spin2/results/`. No uniform spin-two gap has been proved by these calculations.

8.3 A local MPO norm witness and its present limitation

For an MPO core $W^{[j]}_{b,s,t,a}$ in an orthonormal physical basis, let L_b denote the partial operators to its left, with $\|L_b\| \le l_b$. The next partial operator has matrix entries $X^a_{st} = \sum_b W^{[j]}_{b,s,t,a} L_b$. Hence

$$C^a_{st} = \sum_b |W^{[j]}_{b,s,t,a}| l_b, \qquad \|X^a\| \le \|C^a\|_2 \le \sqrt{\|C^a\|_1 \|C^a\|_\infty}.$$

The first inequality follows by bounding the norms of the physical block components of an input vector. Applying this recursively, also from the right, supplies upper bounds at every MPO cut. The calculation uses only local tensors. Rational square-root upper bounds are accepted by integer squaring comparisons.

The Julia implementation `Spin2MPONorms.jl` passes 33 checks, including an exact norm-one result for a 128-site identity MPO and dense controls on random small MPOs. It currently loses too much cancellation: on the four-site ITensors candidates at $\beta = 1$, the resulting bounds exceed 25 although their numerical Hilbert–Schmidt norm is one. These bounds concern the exact quantized candidate tensors; their error relative to the physical thermal operator has not been certified. A tighter norm construction is required before this route can provide the missing thermal input.

8.4 Gauge optimization and certified MPO squaring

A sharper norm witness regards a local core as a matrix from (b,t) to (a,s). Sequential composition bounds the full MPO norm by the product of these local matrix norms. Positive rational diagonal changes of virtual basis cancel exactly on internal bonds. Numerical optimization proposes these gauges and local norm bounds α_j; outward-rounded Arb Cholesky accepts a bound only when $\alpha_j^2 I - W_j^* W_j$ (or the smaller reversed Gram matrix) is certainly positive definite.

For the quantum-number-aware ITensors candidates at $\beta = 1/8$, the four-site product decreases from 3.11769 to a certified bound below 0.16689992, and the eight-site product from 38.97081 to below 0.04945418. Their numerical Hilbert–Schmidt norms are one. At $\beta = 1$, the respective certified products remain above one, so the method still loses too much information there. Saved exact rational tensors, gauges, and local bounds permit independent replay without trusting ITensors or a numerical eigensolver. The norm tests now comprise 68 passing checks.

These local bounds also give a compression certificate without forming a four-layer transfer matrix. Let R_a be the right-tail operators of an MPO, and suppose the coefficient map $v \mapsto \sum_a v_a R_a$ has norm at most γ from Euclidean space to Hilbert–Schmidt space. In a separately gauged copy put $A_a = g_a R_a$, and suppose $\|[A_1 \ \cdots \ A_D]\| \le b$. Then

$$\left\| \sum_{a,b} V_{ab} A_a R_b \right\|_{\mathrm{F}} \le b\gamma \|V\|_{\mathrm{F}}.$$

Indeed, factor this operator into the block row of the A_a and the column of operators $\sum_b V_{ab} R_b$. The latter column has squared Hilbert–Schmidt norm at most $\gamma^2 \sum_{a,b} |V_{ab}|^2$. Both b and γ have

local interval norm witnesses. The additional cap $b \le \gamma(\sum_a g_a^2)^{1/2}$ and optional two-site blocks can sharpen them.

In a left-to-right compression, write the exact local identities as $R_{j-1}P_j = Q_jR_j + E_j$, where P_j is the squared input core and the final step has zero residual. If the preceding Q-prefix map has norm at most c_{j-1} and $\|E_j\|_{\mathrm{F}} \le \eta_j$, telescoping gives

$$\|B^2 - C\|_{\mathrm{F}} \le \sum_j c_{j-1}\eta_j b_{j+1}\gamma_{j+1}.$$

The Julia verifier checks the residuals rationally and all local norm bounds by interval Cholesky. It does not assume exact numerical canonical form or Hermiticity of the approximate MPO. Its 59 tests include independent dense contractions and rejection of corrupted witnesses.

The completed bond-dimension-eight tests at initial $\beta = 1/128$ give relative squaring-error upper bounds 0.020357, 0.053945, and 0.134228 for 8, 16, and 32 sites, respectively. All saved rational witnesses pass independent replay. The denominator here is the output MPO's Hilbert–Schmidt norm. These errors are still too large for the needed repeated squaring, and the initial MPO's distance from the exact Gibbs amplitude remains uncertified. Consequently these advances do not yet establish a thermal seed or a uniform spin-two gap.

Reducing the starting step and allowing the output rank to exceed the input rank is more effective. For the same exact sixteen-site input at $\beta = 1/1024$ and input bond dimension 16, the relative squaring-error bounds are

output dimension	16	24	32	48	64
error upper bound	0.00542631	0.000693457	$1.0764\,10^{-5}$	$1.431\,10^{-6}$	$4.81\,10^{-13}$.

The refinements reuse saved rational input tensors and gauges and raise coefficient-rounding precision from 40 to 48 bits. The dimension-64 certificate has relative error below 4.81×10^{-13} and passes independent replay. A complete sequence of certified squarings, including the initial Gibbs-operator approximation, is still required.

8.5 An exact polynomial starting object

There is a second thermal upper-bound route that avoids a product formula. Put $K = H_N + NI$. The two-site spin-two eigenvalues $-6, -5, -3, 0, 4$ imply $\|K\| \le 5N$. For $m = 2^k$ and $\delta = \beta/(2m)$, choose rational

$$r \ge e^{5N\delta}\frac{(5N\delta)^3}{6}, \qquad Q = I - \delta K + \tfrac{1}{2}\delta^2K^2 + rI.$$

The scalar Taylor remainder and functional calculus give $Q \ge e^{-\delta K} > 0$. Because both are functions of K, their eigenvalues can be raised to the power $2m$, giving

$$Z_N(\beta) \le e^{\beta N}\operatorname{Tr}Q^{2m} = e^{\beta N}\|Q^m\|_{\mathrm{F}}^2.$$

This uses commutation, not a general operator-monotonicity assertion. Furthermore $0 \le Q - e^{-\delta K} \le 2rI$ and $e^{-\delta K} \ge e^{-5N\delta}I$. Hence the exact polynomial upper trace is at most

$$\exp\Bigl(4mre^{5N\delta}\Bigr)\,Z_N(\beta).$$

At $N = 256$, $\beta = 64$, and $k = 28$, outward-rounded scalar arithmetic bounds this factor by 1.000636180. Thus this starting approximation costs less than 0.064%, before any MPO compression. The shifted target would require a certified upper bound on

$$e^{-(a-1)\beta N}\|Q^m\|_{\mathrm{F}}^2.$$

The code `Spin2PolynomialBounds.jl` passes 263 checks, including comparison with the independent exact four-site spectrum. The scalar majorant is established; a rigorous long-chain MPO contraction of its powers is still missing. No spin-two gap follows from the scalar approximation estimate alone.

8.6 Exact compression and the remaining norm bottleneck

The rational polynomial representation and its physical metric are now implemented. In the basis with weights $w = (1, 4, 6, 4, 1)$, a coefficient X_{st} has squared Hilbert–Schmidt weight w_t/w_s. Local norm certificates use rational weighted Gram matrices and outward-rounded Cholesky checks. Bounds are rounded at a relative scale, including for the exponentially small norms of normalized operators.

The initial construction represents K with bond dimension 8 and Q with bond dimension 73. Exact rational row reduction removes dependencies from both ends. For $N = 16$ and $N = 256$ the resulting maximum bond dimension is 34; for $N = 4$ it is 33. This introduces *zero* compression error. The saved witnesses contain local intertwining matrices T_j satisfying

$$W_j T_{j+1} = T_j \widetilde{W}_j, \qquad T_1 = T_{N+1} = 1.$$

These identities telescope to equality of the full operators and have passed independent exact replay at all three lengths. This replaces the earlier dimension-16 starting approximation, whose certified relative error at $N = 256$ was 1.12×10^{-12}, when an exact starting object is preferable.

The first complete short flow uses approximate Gram products $B_j^* B_j$ and an exact spatial reflection after each compression. Reflection preserves operator and Hilbert–Schmidt norms and permits alternating the compression direction without another rounding step. At $N = 16$, four dimension-16 steps, starting from the earlier rounded seed, pass independent replay. They reach only $\beta = 3.814697265625 \times 10^{-6}$. Their forward normalized Hilbert–Schmidt error bound is 9.95×10^{-8}; backward Gram norm bounds reduce it to 8.94×10^{-8}. This is not a useful low-temperature certificate yet.

The exact seed also supplies the rational spectral bound

$$a_0 = \frac{1 + r + 5N\delta + (5N\delta)^2/2}{5^{N/2}} \geq \|Q/5^{N/2}\|.$$

Under the recorded positive normalizations κ_j, reference norm bounds obey $a_{j+1} \geq a_j^2/\kappa_j$. Expanding a Gram-product error around either the exact reference or its approximation permits using the smaller of their separately certified norm bounds in $\epsilon_{j+1} \leq [\delta_j + \epsilon_j(2b_j + \epsilon_j)]/\kappa_j$. For the same four-step run, this reduces the final error further to 4.180×10^{-9}, about 24 times smaller than the original forward bound. The entire source flow was independently replayed before applying this scalar refinement.

An operator-error verifier now bounds the same local compression residuals with independent rational gauges on the output prefix and both input tails. If p_{j-1} bounds the prefix column operator, e_j the gauged residual, and $t_{j+1}^{(1)}, t_{j+1}^{(2)}$ the two tail row operators, telescoping gives

$$\|B^* B - C\| \leq \sum_j p_{j-1} e_j t_{j+1}^{(1)} t_{j+1}^{(2)}.$$

For the first saved sixteen-site step this gives 2.673×10^{-18}, whereas the earlier Hilbert–Schmidt estimate was 8.728×10^{-17}. These bounds control different norms; the operator estimate is intended for the high-entropy part of the flow. It does not by itself bound the final thermal trace.

A checked scalar bridge can connect such an operator estimate to a later trace-moment certificate. If $A \geq 0$ acts on a space of dimension d, $\|A - B\| \leq \eta$, and $\operatorname{Tr}|B|^p \leq M$ for even $p \geq 2$, then for every $t > 0$,

$$\operatorname{Tr} A^p \leq (1+t)^{p-1} M + d(1+1/t)^{p-1} \eta^p.$$

This follows from singular-value perturbation and scalar convexity. For example, if $d = 5^{256}$, $p = 64$, $M \leq 1$, and $\eta \leq 10^{-8}$, choosing $t = 625\eta$ gives an upper factor below 1.0005. The required long-chain operator-error and trace-moment inputs have not yet been certified. The remaining task is still the full low-temperature trace bound, and no uniform spin-two gap has been established.

8.7 A trace correction that retains cancellations

The preceding norm bounds can be wasteful when the desired observable is only the final squared Hilbert–Schmidt norm. Let A be the operator before one compression, let C be its compressed approximation, and suppose $\|A - C\|_{\mathrm{F}} \leq \eta$. The exact identity

$$\|A\|_{\mathrm{F}}^2 = 2 \operatorname{Re} \operatorname{Tr}(C^* A) - \|C\|_{\mathrm{F}}^2 + \|A - C\|_{\mathrm{F}}^2$$

gives the upper certificate

$$\|A\|_{\mathrm{F}}^2 \leq 2 \operatorname{Re} \operatorname{Tr}(C^* A) - \|C\|_{\mathrm{F}}^2 + \eta^2.$$

An upper enclosure of the signed overlap and a lower enclosure of $\|C\|_{\mathrm{F}}^2$ suffice. The remaining residual contribution is quadratic, whereas the triangle certificate $(\|C\|_{\mathrm{F}} + \eta)^2$ has a linear error term. For a squaring $A = B^2$ (or a Gram product $B^* B$), the overlap has three MPO layers, rather than the four layers of the direct squared norm. This changes the contraction cost and the certificate simultaneously.

The script `trace_correction_probe.jl` replays the rational local residual certificates and encloses the overlaps with ball arithmetic. Four-site checks against a direct uncompressed contraction give:

Output dimension	Triangle upper/reference	Corrected upper/reference
16	1.155853810	1.004971213
32	1.013744177	1.000046216

At dimension 32, the excess upper bound is reduced about 297-fold without changing the compressed candidate. These are certificates for *one compression of a saved candidate operator*. They do not certify its initial Gibbs error or earlier squarings. The following extension retains these earlier contributions as well.

8.8 A backward certificate for the complete trace flow

Use the real Hilbert–Schmidt pairing $\langle X, Y \rangle = \operatorname{Re} \operatorname{Tr}(X^* Y)$. Let S_j be either the identity or spatial reflection, and set

$$A_{j+1} = S_j(A_j^* A_j)/\kappa_j, \qquad D_j = A_j - B_j, \qquad R_j = S_j(B_j^* B_j)/\kappa_j - B_{j+1},$$

where $\kappa_j > 0$. No Hermiticity of the approximations B_j is required. Choose backward approximations G_j with $G_K = B_K$, and define

$$T_j = \frac{B_j\big(S_j(G_{j+1}) + S_j(G_{j+1})^*\big)}{\kappa_j} - G_j.$$

Suppose separately verified bounds give $\|D_j\|_{\mathrm{F}} \le \epsilon_j$, $\|T_j\|_{\mathrm{F}} \le \tau_j$, and $\|G_{j+1}\| \le \gamma_j$. Define

$$c = \|B_K\|_{\mathrm{F}}^2 + 2\langle G_0, A_0 - B_0\rangle + 2\sum_{j=0}^{K-1} \langle G_{j+1}, R_j\rangle,$$

$$r = 2\sum_{j=0}^{K-1} \left(\tau_j \epsilon_j + \frac{\gamma_j \epsilon_j^2}{\kappa_j}\right).$$

Then

$$\boxed{c - r \;\le\; \|A_K\|_{\mathrm{F}}^2 \;\le\; c + r + \epsilon_K^2.}$$

For interval contractions, replace c by its lower or upper endpoint on the corresponding side.

Indeed, expanding the Gram step gives

$$D_{j+1} = R_j + \frac{S_j(B_j^* D_j + D_j^* B_j + D_j^* D_j)}{\kappa_j}.$$

Pairing with G_{j+1} transfers the linear part to $G_j + T_j$. Telescoping and expanding $\|B_K + D_K\|_{\mathrm{F}}^2$ yield

$$\|A_K\|_{\mathrm{F}}^2 = c + \|D_K\|_{\mathrm{F}}^2 + 2\sum_{j=0}^{K-1} \left(\langle T_j, D_j\rangle + \frac{\langle G_{j+1}, S_j(D_j^* D_j)\rangle}{\kappa_j}\right).$$

Cauchy–Schwarz bounds the first remainder; operator/trace-norm duality bounds the second by $\gamma_j \epsilon_j^2/\kappa_j$. Thus the signed linear errors are retained, and the remaining errors are products of primal and backward errors or squares of primal errors.

The backward operator norms need not be replaced by Hilbert–Schmidt norms. If $b_j \ge \|B_j\|$ and $g_K \ge \|B_K\|$, the same dual defects give

$$g_j = \frac{2 b_j g_{j+1}}{\kappa_j} + \tau_j \;\ge\; \|G_j\|.$$

At each step one may take the smaller of this bound and a separately certified Hilbert–Schmidt norm. This avoids an unnecessary dimension-dependent loss when the thermal operator is close to a multiple of the identity.

The complete five-step, four-site benchmark in `adjoint_trace_probe.jl` reaches $\beta = 4$. The following ratios use an independently enclosed uncompressed Trotter trace:

Forward rank	Backward intermediate rank	Triangle/reference	Corrected/reference
358	625	1.232375999	1.018555350
564	625	1.008410682	1.000024309
564	358	1.008410682	1.000301094

The second row improves the excess upper bound about 346-fold. The third compresses the backward intermediates and retains about 28-fold improvement; its terminal $G_K = B_K$ still has rank 564. The saved rational forward and backward residuals, positive-vector operator-norm witnesses, and signed contractions all pass independent replay. These are full-flow enclosures of a four-site Trotter trace. Its upper endpoint bounds the Gibbs trace, but its lower endpoint does not supply a Gibbs lower bound.

Mixed MPO products and streamed three-layer overlaps are now implemented with local rational witnesses. The latter avoid explicitly constructing product cores or a full physical operator. A

first sixteen-site single-step check reduces the upper bound by 6.8181×10^{-11} times the candidate squared norm. This is a comparison of two certified upper bounds, not a measurement of their error against the exact trace.

The complete backward correction is now also implemented for the saved four-step, sixteen-site MPO flow, including the initial seed error. The forward and backward maximum bond dimensions are both 16. The first unconditioned backward construction was weaker than the triangle bound. Reusing the forward rational gauges in the reflected direction substantially reduces its certified defects. Replacing reciprocal gauge proposals by nearby dyadics reduces arithmetic cost while preserving exact cancellation of the gauges in the represented operator.

After independently replaying all forward and backward witnesses, and applying the backward operator-norm refinement, the ratios are

$$\frac{U_{\text{triangle}}}{\|B_K\|_{\mathrm{F}}^2} \simeq 1.0000000083597809, \qquad \frac{U_{\text{corrected}}}{\|B_K\|_{\mathrm{F}}^2} \simeq 1.0000000000036466.$$

With $L_{\text{corrected}}$ the certified lower endpoint for the exact polynomial trace, the interval-width comparison is

$$\frac{U_{\text{triangle}} - L_{\text{corrected}}}{U_{\text{corrected}} - L_{\text{corrected}}} \simeq 1307.0, \qquad \frac{U_{\text{corrected}}}{L_{\text{corrected}}} - 1 < 6.399 \times 10^{-12}.$$

These compare certified enclosures, not errors relative to a known exact trace. The portable result is `mpo-adjoint-flow-refined-n16.json`. Its verifier reconstructs the backward inputs from the saved forward candidates, checks local rational product residuals and norm bounds, and recomputes every signed overlap. The resulting Gibbs upper bound is still at $\beta = 1/262144$, leaving 24 further doublings to reach $\beta = 64$.

Longer error recurrences now support upward dyadic rounding after each step. Each rounded value is checked to dominate the exact rational recurrence, with 128 relative binary digits retained. This prevents denominator growth from repeatedly squaring exact error fractions in an otherwise bounded recurrence. A thousand-step stability check keeps denominators at most 150 binary digits; an actual two-step four-site flow with rounded errors also passes independent replay.

There is a further useful improvement to propagation. If independently certified bounds give $\|A\| \le a$, $\|B\| \le b$, and $\|A - B\|_{\mathrm{F}} \le \epsilon$, then

$$\|A^*A - B^*B\|_{\mathrm{F}} \le \epsilon \min\{2a + \epsilon,\ 2b + \epsilon,\ a + b\}.$$

The last coefficient follows from $A^*A - B^*B = A^*(A-B) + (A^*-B^*)B$. Using both norm certificates avoids an unnecessary ϵ^2/κ term when the normalized operator norms are much smaller than the available Hilbert–Schmidt error. For example, the conditional inputs $a = 5^{-128}$, $b = 2a$, $\kappa = a$, zero local residual, and $\epsilon = 10^{-12}$ give propagated error 3×10^{-12}, whereas the earlier coefficient gives more than 10^{60}. This is a conditional propagation stress test; it supplies no local compression residual. The two-norm formula is implemented in both the generator and verifier. The required $N = 256$, $\beta = 64$ thermal certificate remains missing, and no uniform spin-two gap has been proved.

8.9 Balancing full auxiliary matrices at 256 sites

Diagonal auxiliary scalings do not resolve the norm problem for the saved dimension-16 starting MPO at $N = 256$: the first numerical search gave a product of local norm estimates about 1.659×10^{75} times the analytic bound a_0 for the exact seed. Full matrix gauges give a substantially different result. A stable QR/SVD sweep computes left-prefix Schmidt data; the proposal splits each Schmidt scale between its left and right factors and rotates the auxiliary basis accordingly.

The proposed matrices are rounded to rational matrices G_j. Their inverses and the transformed cores are computed exactly:

$$\widetilde{W}_j = G_j W_j G_{j+1}^{-1}, \qquad G_1 = G_{N+1} = 1.$$

These factors cancel on every internal bond, so the global operator is unchanged. Verified local operator bounds then give

$$\|B_0\| \le \prod_j \|\widetilde{W}_j\|_{\mathrm{local}} < 1.000000701\, a_0, \qquad a_0 = \frac{1 + r + 5N\delta + (5N\delta)^2/2}{5^{N/2}}.$$

For the actual saved $N = 256$ candidate, the certified upper norm is approximately $3.403345322 \times 10^{-90}$. The proof is stored in `schmidt-seed-norm-n256.json`. Independent replay checks its source fingerprint and physical adjoint, exact gauge inverses, and weighted local Gram inequalities. The numerical Schmidt decomposition is only a proposal and is not trusted by the verifier.

Exact initial seeds can also be conditioned with rational matrix gauges. At $N = 4$, the conditioned seed retains zero initial error; two subsequent dimension-32 squarings have total certified error below 1.225×10^{-13} and pass independent replay. The earlier raw exact-basis first-step error bound exceeded 7358, illustrating why an exact representation alone is insufficient for effective truncation.

The full-matrix norm certificate resolves one obstruction at the target chain length. It does not certify the compression errors of the complete thermal flow. The separate operator-error residual routine currently supports matrix gauges, as do the Hilbert–Schmidt Gram and adjoint routines. The $\beta = 64$ trace and the uniform spin-two gap remain unproved.

Normalization without a long interval contraction. The first matrix-gauge squaring attempt passed its local residual checks but failed in the final 192-bit global norm contraction. For the left-compressed output C, let α_j be the verified norm bound of its jth left-prefix map and let T be the terminal core. Then

$$\|C\|_{\mathrm{F}} \le \|T\|_{\mathrm{F}} \prod_{j=1}^{N-1} \alpha_j.$$

The terminal norm uses the rational physical metric, and its upper square root is checked by exact integer arithmetic. This gives a normalization bound without propagating a dense interval environment through all sites. Any positive rational normalization is allowed: its exact scale is retained in the final trace formula. At four sites, the resumed three-step flow using this bound passes independent replay, with total error below 1.326×10^{-11} at $\beta = 1/524288$.

Interior bridge products can also be enclosed by interval arithmetic during candidate generation. The rounded candidates remain exact rational arrays, and each residual bound includes the interval enclosure. Small-system tests pass both interval and exact-rational replay, including zero and extremely small operators. The last core is still contracted exactly. A raw local witness is now saved before normalization so that a later failure does not discard the expensive squaring computation. These changes do not supply the missing low-temperature trace.

With the revised normalization and interval bridge contractions, the first dimension-16 squaring at $N = 256$ completes at $\beta = 1/2097152$. Its propagated Hilbert–Schmidt error is below 6.762×10^{-8}, with amplification coefficient below 2.001. The square and its local verification take approximately 157 seconds. The earlier diagonal-gauge run at the same length and dimension gave an unusable error bound above 7.60×10^{25}; the tail grouping and normalization also differ, so this is not an isolated gauge comparison. The result and portable witness are saved as

`polynomial-schmidt-flow-n256.json` and its associated step file. This completes only one of the 28 squarings: 27 further temperature doublings and their error certificates are still required.

8.10 Certifying the benefit of local orthogonality

The sum of local truncation errors can be replaced by a backward quadratic recurrence that checks the cross terms. At cut j, write the local compression as $M_j = U_j R_j + E_j$, and let W_j be its physical row metric. In addition to the existing bounds $\|E_j\|_{\mathrm{F},W_j} \le \delta_j$ and $\|U_j\| \le \alpha_j$, certify

$$\|U_j^{\mathsf{T}} W_j E_j\|_{\mathrm{F}} \le \theta_j.$$

Let Γ_{j+1} bound the remaining product-tail coefficient map. If e_{j+1} bounds the suffix compression error, then

$$\|D_j\|_{\mathrm{F}}^2 \le (\delta_j \Gamma_{j+1})^2 + \alpha_j^2 e_{j+1}^2 + 2\theta_j \Gamma_{j+1} e_{j+1}, \qquad e_N = 0.$$

The quantity D_j is the actual suffix error; e_j is chosen as an outward-rounded upper square root of the right-hand side. To prove the estimate, decompose the suffix error as $E_j T_{j+1} + U_j D_{j+1}$ and move U_j across the weighted inner product in the cross term. The two factors are bounded by $\theta_j \Gamma_{j+1}$ and e_{j+1}.

For exact orthogonal projections $\theta_j = 0$, recovering the usual sum-of-squares benefit. Rounded singular vectors need not be orthogonal: the stored θ_j bounds their actual defects, checked directly in rational or outward-rounded interval arithmetic. One may also use $\theta_j \le \alpha_j \delta_j$ and retain the previous triangle bound whenever it is smaller after rounding. The implementation can keep all saved cores and bridges fixed while recomputing this sharper certificate. Small-system tests compare against the exact physical operator and reject a corrupted projection-defect bound.

At $N = 256$ and $\beta = 1/2097152$, the following first-step error bounds all pass independent replay:

Error estimate	Retained dimension	Propagated error upper bound
Triangle sum	16	6.762×10^{-8}
Projection-defect recurrence	16	4.259×10^{-9}
Triangle sum	32	8.361×10^{-11}
Projection and rescaled gauges	16	7.740×10^{-11}
Projection and fitted bridges	32	7.413×10^{-12}

The first two rows use exactly the same compressed cores and bridges; the improvement of approximately 15.9 is due to the certificate. The third row instead increases the retained dimension, improving the original propagated bound by approximately 809. The rescaled dimension-16 row retains exactly the original compressed cores and normalization, checked directly in the saved witnesses, and improves the original bound by more than 870. The final row uses 64-bit relative quantization and fits each outgoing bridge to the actual rounded left factor before certifying the residual and its projection defect. Its improvement exceeds 9000 relative to the original dimension-16 bound. These comparisons concern the first thermal-step error; they do not establish the required low-temperature trace.

Auxiliary gauges can also be optimized for the compression error, separately from the candidate operator norm. If $G'_j = D_j G_j$ for positive rational diagonal matrices D_j, replace the saved bridges by $R'_{kab} = R_{kab}/(D_{j+1})_{aa}$ on the first input's auxiliary leg. The diagonal factors cancel in the local products, so every compressed output core remains unchanged, including the exact terminal core. The residual and tail bounds must be recomputed for the new gauges. This change of certificate

is implemented and tested with exact rational replay; a numerical gauge search is only a proposal for these checks. The successful dimension-16 rescaling uses row factors corresponding to Schmidt power 0.2, rather than 0.5. It permits a looser candidate operator-norm bound, about $2.356a_0$, in exchange for smaller certified compression errors. The total propagated bound, including the resulting larger amplification of the initial error, is the fourth row above.

Reducing the arithmetic and storage burden. Matrix gauges and their inverses are still checked exactly over the rationals. Their interior transformed cores may now be evaluated as outward-rounded interval enclosures, avoiding storage of every large rational coefficient created by a dense inverse. The terminal core is kept exact. If an interval residual check is inconclusive, the verifier recomputes that core in rational arithmetic. Exact replay remains available, and the new path independently verifies the rescaled 256-site certificate.

Reflected outputs are now stored by their exact reconstruction rule $B_{j+1} = \mathcal{S}(C_j)/\kappa_j$, avoiding a duplicate copy of every coefficient. The four-site witness shrinks from 4,035,669 to 2,800,932 bytes; interval and exact-rational replay reconstruct the same final candidate as the older format. The improved first-step certificates are saved in `polynomial-projection-rescaled-d16-n256.json` and `polynomial-projection-d32-fitted-prefix1-n256.json`. The hybrid trajectory has completed two independently replayed doublings, reaching $\beta = 1/1048576$ with propagated error at most $1.747656072 \times 10^{-10}$. The target $\beta = 64$ still requires 26 further doublings. No uniform spin-two gap has been certified.

9 Higher-order initialization by positive polynomial squares

The number of subsequent squarings can be reduced by increasing the degree of the initial polynomial. Write $K = H + NI$, $M = 5N$, $m = 2^p$, and $\delta = \beta/(2m)$. For any positive integer d, choose a rational

$$r \geq e^{\delta M} \frac{(\delta M)^{d+1}}{(d+1)!}, \qquad Q = \sum_{j=0}^{d} \frac{(-\delta K)^j}{j!} + rI.$$

Taylor's remainder and functional calculus give

$$e^{-\delta K} \leq Q \leq e^{-\delta K} + 2rI \leq (1 + 2re^{\delta M})e^{-\delta K}.$$

These operators commute, so

$$Z_H(\beta) \leq e^{\beta N} \operatorname{Tr} Q^{2m} \leq \exp(4mre^{\delta M}) Z_H(\beta).$$

The following scalar bounds at $N = 256$, $\beta = 64$ have been checked with outward-rounded arithmetic; displayed upper bounds are rounded upward.

Degree d	Subsequent squarings p	Inflation upper bound
2	28	1.000636180
4	21	1.000206619
6	18	1.000646796
8	17	1.000076720
12	16	1.000000327

These bounds concern the exact polynomial; an MPO approximation still requires its own certificate. The recorded calculation is `polynomial-degree-screen-n256.json`.

For the fourth-order polynomial there is an exact rational identity

$$P_4(x) = \left(1 + \frac{x}{2} + \frac{x^2}{12}\right)^2 + \frac{1}{12}\left(x + \frac{x^2}{2}\right)^2 + \frac{1}{72}(x^2)^2.$$

It permits construction from three quadratic Gram certificates. This is systematic: for $P_{2k}(x) = \sum_{j=0}^{2k} x^j/j!$,

$$P_{2k}(x) = \frac{1}{(2k)!}\int_0^\infty e^{-t}(t+x)^{2k}\,dt.$$

The moment Gram matrix of $(t+x)^k$ is rational and positive definite. An exact LDL^T decomposition gives positive rational weights times squares of degree-k polynomials. The implementation checks the coefficient identity exactly, including examples through degree 32.

To track normalization, put $\sigma = 5^{-N/2}$ and $A_i = \sigma p_i(-\delta K)$ for the three quadratic factors above. They are self-adjoint, and

$$\sigma Q = \sigma^{-1}\sum_i w_i A_i^* A_i + r\sigma I.$$

If $\|C_i - A_i^* A_i\|_{\mathrm{HS}} \le \eta_i$ and a final compressed candidate B differs from $\sigma^{-1}\sum_i w_i C_i + r\sigma I$ by at most ρ, then

$$\|B - \sigma Q\|_{\mathrm{HS}} \le \rho + \sigma^{-1}\sum_i w_i \eta_i.$$

The Julia seed builder uses the existing Gram verifier for each term and a separate contraction for the final compression error. This trades three initial Gram constructions for seven fewer subsequent squarings. A net runtime improvement and a 256-site fourth-order seed certificate have not yet been established. The thermal-flow generator and verifier now accept this seed format and replay all three factor witnesses. The four-site seed and two subsequent squarings now pass independent replay. The first four-site prototype, at $\beta = 4$, $p = 8$, and retained dimension 32, gives a generator-checked initial error bound of 4.562×10^{-4}. Only 1.322×10^{-7} comes from the final compression; the factor Gram certificates dominate. A separate four-site test uses the target initial step $\beta = 64$, $p = 21$, and retains up to 34 components in the quadratic factors. With Gram output dimension 32, it gives a generator-checked bound of 2.072×10^{-8}, with less than 1.759×10^{-15} from final compression. Retaining dimension 34 gives 1.815×10^{-9} and passes independent replay.

Rescaling the three factor gauges, with powers 0.05, 0.05, and 0.25, gives an independently replayed bound of 3.999×10^{-13}. Exact comparisons of the saved files confirm unchanged inputs, factor outputs, and final candidate: the bound improves by more than 4500 without changing the trial operator. The numerical screen proposes the rescalings; each selected bound is recomputed with rational and interval arithmetic. The result is `quartic-seed-d34-refined-n4.json`.

After two further squarings, at $\beta = 1/8192$, the default gauges give an independently replayed error bound of 3.645×10^{-8}. Optimizing the residual gauges at each step reduces the independently replayed bound to 2.166×10^{-11}, an improvement of more than 1680. Both input and output cores, and both normalization constants, remain exactly unchanged. These are four-site, high-temperature tests; they do not establish the required low-temperature trace bound at $N = 256$.

An additional variant retains exact quadratic inputs. Two rational compression sweeps produce local intertwining identities, which the verifier checks before accepting zero input error. This addresses a large-system difficulty: the normalized factor's operator norm is of order $5^{-N/2}$, and

an ordinary absolute Hilbert–Schmidt rounding error can greatly exceed this scale. Removing that input error avoids its amplification in the Gram estimate. Conditioning is essential: the raw four-site exact representation gives an error bound above 1.2×10^4. Exact right conditioning lowers it to 1.254×10^{-9}, and subsequent gauge refinement gives 3.410×10^{-13}, with all three factor input errors exactly zero. The refined candidate and factor outputs are unchanged. This exact-input variant now passes independent replay.

Lossless witness encoding. Large dyadic denominators can be stored by their binary exponents, without altering any coefficient. On the existing three-step four-site fixture, this reduces the combined witness size from 7,562,803 to 5,768,539 bytes, or 23.7%. Every decoded field agrees exactly with its predecessor, and the reencoded flow passes independent replay. Legacy rational encodings remain readable. This is a storage improvement; an end-to-end runtime improvement has not been measured.

Separate gauges for separate inequalities. The complete-operator norm and the compression residual need not use the same auxiliary gauges. If a second family F_j, with identity boundary gauges, gives independently verified local operator bounds v_j, then $b_F = \prod_j v_j$ bounds the complete input operator. Hence the Gram error coefficient may use $b = \min\{b_F, b_G, b_{\mathrm{HS}}\}$, while retaining the original residual-tail bounds from the family G_j. In particular, the global bound b_F is never substituted for a suffix-map bound. The verifier supports this additional certificate and checks its gauge cancellations and local inequalities. Explicit small-operator tests pass in exact and interval arithmetic and reject corrupted auxiliary bounds. The four-step integration run reaches $\beta = 1/2048$ with an independently replayed Hilbert–Schmidt error bound of 1.311×10^{-9}. An additional 256-site improvement from this separation has not yet been demonstrated.

Full matrix gauges for operator errors. The operator-error residual certificate now handles full matrix gauges. Let G_j be the original first-input gauge, F_j, S_j the new tail gauges, and P_j the new output-prefix gauge. The exact residual transforms by

$$E'_{q'sta'b'} = \sum_{q,a,b} (P_j)_{q'q} E_{qstab} (G_{j+1} F_{j+1}^{-1})_{aa'} (S_{j+1}^{-1})_{bb'}.$$

The prefix and tail gauge factors cancel in the full telescoping identity. Three separate auxiliary-axis contractions implement this formula without forming a Kronecker matrix. The checker visits one residual at a time; interval arithmetic has an exact local fallback. All 138 tests pass, including exact reconstruction of the complete residual on small systems with noncommuting gauges, unequal auxiliary dimensions, and rejection of corrupted bounds.

On the second four-site quartic step, the initial operator-error bound is 9.762×10^{-8}. Selecting independent gauge powers $(0.1, 0.05, 0)$ reduces it to 2.834×10^{-12}, an improvement exceeding 34,000, with input, output, and original bridges fixed. Both bounds pass independent replay. The optimized operator bound is still looser than the saved Hilbert–Schmidt bound 6.613×10^{-13} on this small example.

Operator error at initialization. The unequal-dimension extension certifies linear compression as $A \cdot I$. Starting from a verified exact quadratic polynomial, four-site compression to dimension 16 gives an operator-error bound 1.467×10^{-7}. Changing only the error gauges lowers it to 2.280×10^{-13}, with the full product witness and candidate unchanged; independent replay passes. Retaining dimension 34 and refining its gauges gives an independently replayed bound 7.289×10^{-17},

or 1.823×10^{-15} relative to the analytic operator-norm bound. The long-chain operator-error and trace-moment inputs are still missing.

Backward Schatten propagation. A systematic alternative to forward error propagation uses a different Schatten norm at each doubling. Let $D = 5^N$, let k be the number of Gram compressions, and put $p_j = 2^{k+1-j}$. Suppose C_j approximates $B_{j-1}^\dagger B_{j-1}$ with certified Hilbert–Schmidt and operator errors ζ_j and σ_j, respectively, and B_j equals C_j/κ_j, up to a unitary spatial reflection, with $\kappa_j > 0$. Starting from $m_k \geq \|B_k\|_2$, define, with upward rounding,

$$m_{j-1} = \sqrt{\kappa_j m_j + \min\{\zeta_j, D^{1/p_j}\sigma_j\}}.$$

The Schatten triangle inequality and $\|B^\dagger B\|_p = \|B\|_{2p}^2$ prove inductively that $m_j \geq \|B_j\|_{p_j}$. An unavailable operator certificate is omitted from the minimum. If $A_0 = Q/\sqrt{D}$ is the exact positive polynomial seed and $A_0 - B_0$ has certified errors η_2, η_∞, then

$$\operatorname{Tr} Q^{p_0} \leq D^{p_0/2}\big(m_0 + \min\{\eta_2, D^{1/p_0}\eta_\infty\}\big)^{p_0}.$$

For $Q \geq e^{-\delta(H+NI)}$, the two operators commute, so at $\beta = \delta p_0$ this gives a shifted thermal trace upper bound after multiplication by $e^{-(a-1)N\beta}$.

This recurrence avoids the forward global-operator amplification factors. At early steps D^{1/p_j} is close to one; at late steps the minimum can use the Hilbert–Schmidt error. The implementation checks every rounded square root by an exact rational squaring inequality and obtains dimension factors by repeated square roots. All 107 additional tests pass, including explicit matrix chains, non-Hermitian candidates, reflections, and the exponent required for 28 squarings at $N = 256$. A replay script connects these scalar inequalities to saved MPO witnesses. The required low-temperature long-chain inputs remain unproved; this recurrence is not itself a spin-2 gap certificate.